and filters out all non-human detections.

This step ensures that only the pedestrian class is sent to the next step, which is tracking. The SORT algorithm now handles the functioning as the system's short-term memory companion. It assigns a unique ID to all the detected pedestrian class objects and maintains this identity across all the frames, even if the individuals in the frame try to disappear or overlap with other objects. This temporal continuity allows the system to follow the path of each pedestrian even if they move through the frame, just like a normal human driver should be tracking someone walking across a street while driving.

After analysing each frame, the AI system overlays bounding boxes and ID tags onto the video in the system, providing a more clear and persistent view of the pedestrian movement across the frames. This output video is then compiled into a new annotated video format that visualises detection and has an intelligent tracking component.

This pipeline is truly exceptional because of its real-time capability; it sees the pedestrians and remembers them, as well as reacts to their movements quickly. This makes the pipeline adequate for deployment in real live systems, and can be integrated into autonomous vehicles or used for smart surveillance purposes. Ultimately, this system works like a human driver. Its real-world use cases include autonomous vehicles, urban surveillance, smart crosswalks and fleet monitoring.

Edge performance and optimisations

Performance is non-negotiable in this project. Real-time systems in the industry can't afford lag, i.e., high latency.

To ensure a very smooth 30 FPS playback, YOLOv5s (the smaller but faster version of the model) has been used. The computation has been offloaded to the GPU that used torch.device('cuda') where available. Frame size has been trimmed and used at almost half-precision (that is float16) inference.

But for environments that don't have GPU access, this model still works on a CPU, albeit at a lower frame rate.

Real-world challenges

There are real-world challenges when we start integrating such AI systems in real cars running on the road with uncontrolled environmental conditions.

Lighting: Early morning shadows and low-light environments are both hurdles. In this project, this challenge was handled by a preprocessing step that used histogram equalisation and contrast adjustment techniques.

Occlusion: This occurs when people overlap or cross paths. The major problem here is that their IDs may get switched. The SORT algorithm can handle mild occlusions but falters when faced with some complex crowd movement scenarios. So, in future updates, there are plans to incorporate the DeepSORT algorithm, which adds appearance-based embedding for higher accuracy.

From Python script to AV module

What makes this project more than just a coding experiment is its extensibility. You can integrate it into a larger autonomy stack. Here's how:

Component	Integration
YOLOv5	Detection backbone
SORT	Realtime multi-object tracking
ROS / WebRTC	Live streaming into navigation stack
TensorRT	Edge deployment for NVIDIA Jetson or Orin
Flask API	Serves frame results to front-end dashboards

Remember, you're not building an AI toy—you're contributing to the digital senses of a vehicle. This project is special not because of the maths behind it but the human instinct being replicated using an AI vision system. When a 6-year-old starts crossing a zebra line, a human driver slows down. This system allows a machine to do the same, not because it was told to, but because it learned to observe, identify, and track the human in its path.

The last step before deployment

You've got a working model. Now what?

- I recommend you test it on multiple city datasets (e.g., CityPersons, ETHZ, etc).
- Further finetune it for local traffic behaviour as well.
- You can also add alert mechanisms, for either audio or in-vehicle heads-up display.
- For more advancement you can integrate it with braking systems or steering logic.

And that's the journey so far...

You don't need to be Tesla or Waymo to contribute to safer streets. Sometimes, all you need is a laptop, a good model, and a genuine concern for human life.

This project is my step in that direction—and, perhaps, it can be yours too. END

By: Abhishek Raj Permani

The author is a distinguished data scientist and AI researcher, with a master's degree in data science and AI. He has worked in companies like Jaguar Land Rover and Samsung Research, and is passionate about solving real-world problems in areas such as autonomous systems, transformer models, and recommendation engines. He frequently shares his knowledge at national conferences and forums.